

OSVVM's Co-simulation Framework

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1 Overview

The co-simulation features of OSVVM are a compliment to the Address Bus Model Independent Transactions features, as outlined in [1] and the Streaming Model Independent features (see [3]). It is highly recommended that these documents are read before this one, as it will be assumed that the reader is familiar with the transactions associated with that feature.

A typical address bus transaction test bench might look something like the diagram below for an AXI4 environment, which is based on the example test bench in `AXI4/Axi4/testbench/TbAxi4Memory.vhd`. with the `Axi4Memory` VC standing in for a unit under test component.

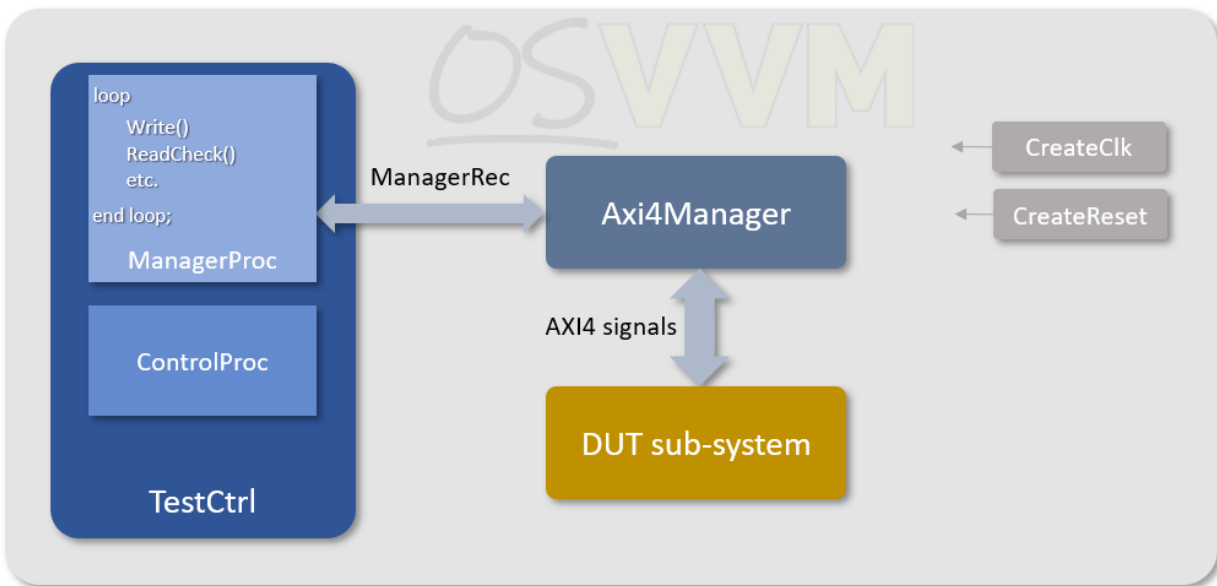


Figure 1: Address Bus Transaction Test Environment

In this example the test has a manger process using calls to the address bus procedures, within an operation loop, to generate transaction via the `ManagerRec` which the `Axi4Manager` VC component translates into specific AXI4 signalling. In this case the target is another OSVVM VC, which is a memory component but would normally be the DUT component or sub-system. A subordinate process inspects arriving traffic and performs checks. In some scenarios the subordinate process generates the responses but, in this case, the `Axi4Memory` VC generates these internally. The advantage of this system is that the transactions are abstracted away from the bus specific model with re-usable components. If the `Axi4Manager` VC is replaced with another address bus VC, then the manager code remains unchanged and can drive the new system

The aim of the co-simulation features is to extend this architecture (and similarly for streaming) to allow the generation of the transactions to be performed within a C/C++ domain. It extends the calls to the address bus procedures to calls to methods within a C++ class. This opens up new possibilities not available in a pure VHDL domains. Some of these are listed below:

- Writing of tests in C++

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- Test development environment extended to software engineers
- Running C++ models and code on simulated hardware
 - Instruction set simulators or cycle accurate processor models
 - System models with a mix of software models and logic IP
- Connection to an external program via TCP/IP sockets
 - Where model might not be callable directly from the C++ code

With co-simulation of processor and system models, the possibility of developing software alongside the logic IP exists before silicon is available, but using both that actual hard and soft components together and not facsimiles, reducing risk.

The co-simulation features replace the transaction calls to TestCtrl's ManagerProc with a connection to the co-simulation interface. The diagram below shows the updated framework for address bus model independent AXI4 manager VC.

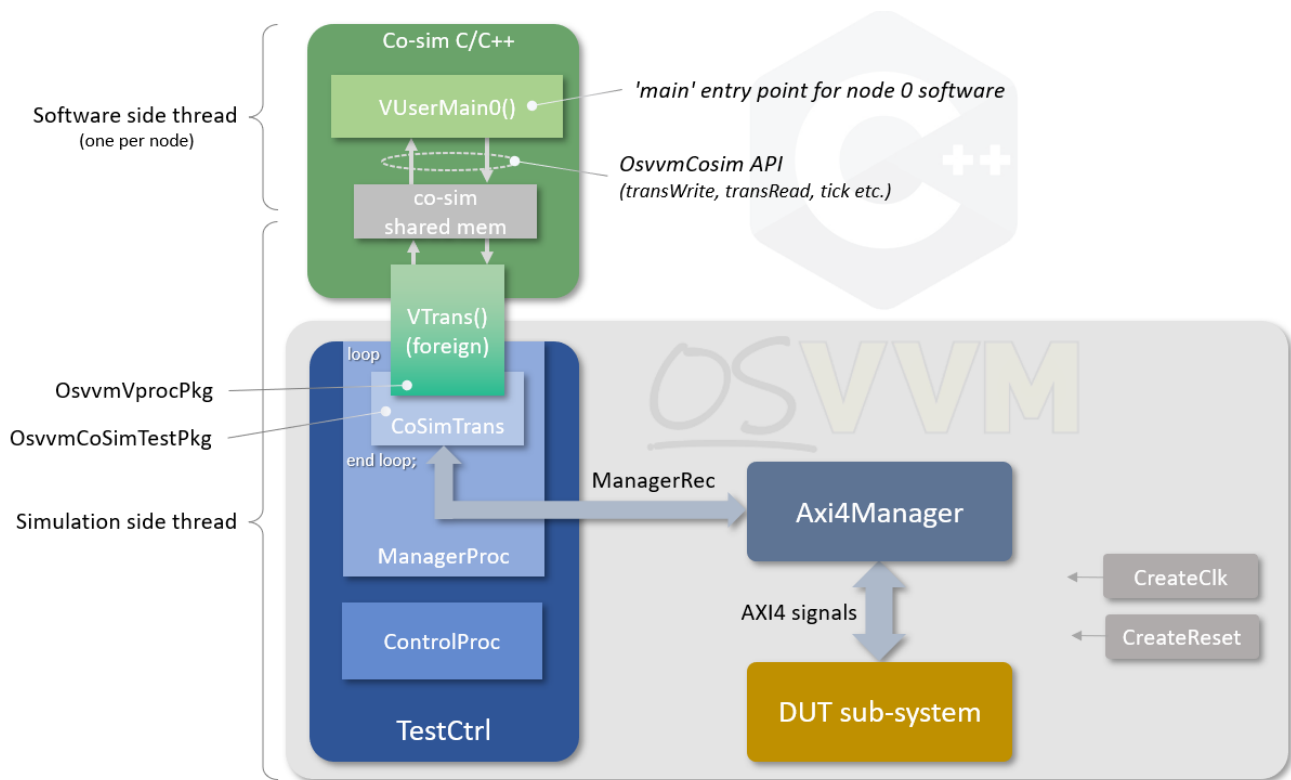


Figure 2: Co-simulation Environment Address Bus Example

In place of calls directly to the address bus transaction procedures, a **CoSimTrans** procedure is provided to replace these and drive **ManagerRec** instead. This hides details away for calls to the underlying co-simulation code, with calls to foreign procedures that communicate with user program. The code, at startup will call a function (**VUserMain0** for 'node 0', for example) that the user provides with all their code and sub-code called from there (much like **main()**). The user code now has access to the transaction API provided by the co-simulation code, and can generate read and write transactions in the OSVVM domain.

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A complementary setup for subordinate responses is also available. Here the co-simulation features replace the transaction calls to TestCtrl's SubordinateProc with a connection to the co-simulation interface. The diagram below shows the updated framework for address bus model independent AXI4 subordinate VC.

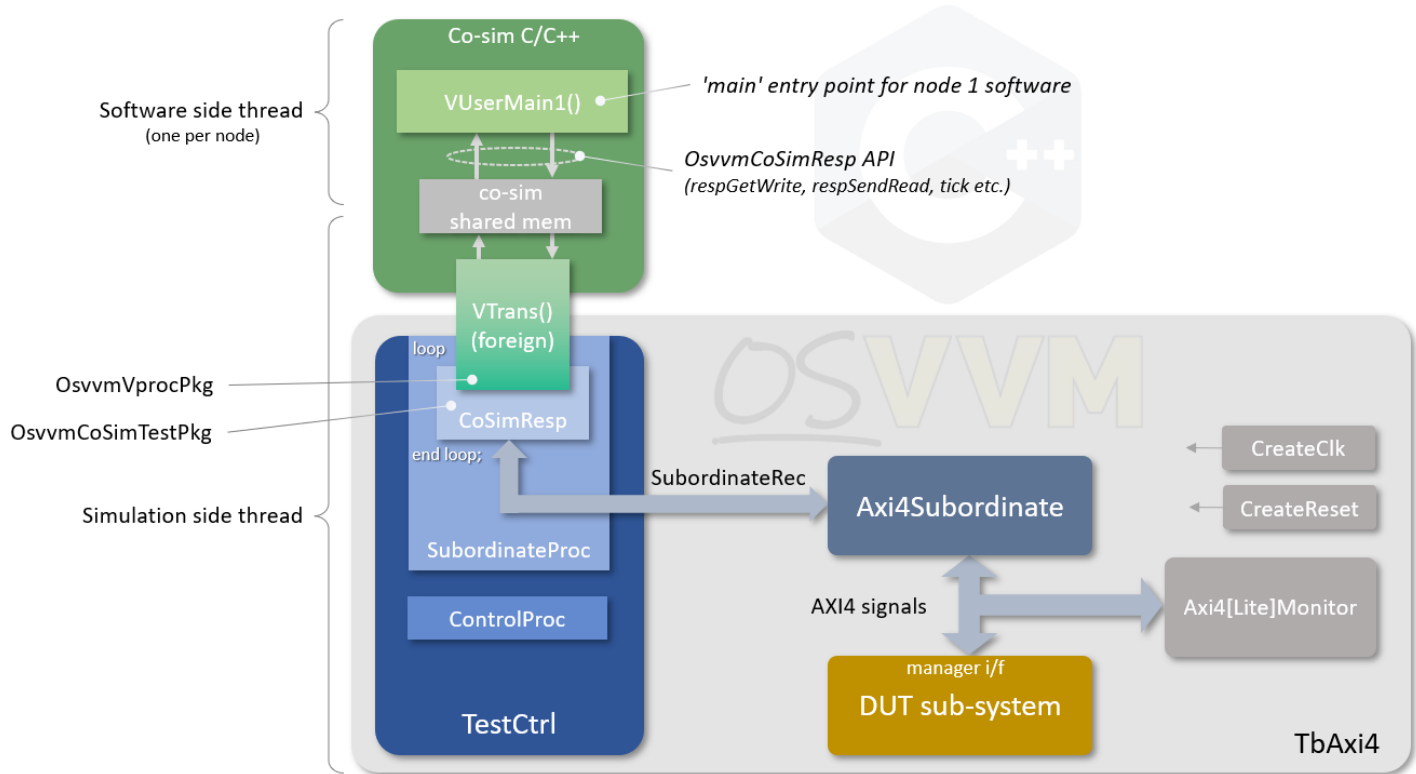


Figure 3: Co-simulation Environment Address Bus Responder Example

In place of calls directly to the address bus transaction procedures, a CoSimTransResp procedure is provided to replace these and drive SubordinateRec instead. This hides details away for calls to the underlying co-simulation code, with calls to foreign procedures that communicate with user program. The code, at startup will call a function (VUserMain1 for 'node 1', for example) that the user provides with all their code and sub-code called from there. The user code now has access to the transaction API provided by the co-simulation code, and can send read responses and fetch received write transactions in the OSVVM domain. The complexity of the user code to model other system features is now unlimited, and can direct address bus accesses towards the simulation, and components modelled there via the API as appropriate.

A similar setup is found for the stream model independent setup. The diagram below shows an example test setup using the xMii MAC VC for Ethernet (with a setup for the xMii PHY VC being very similar):

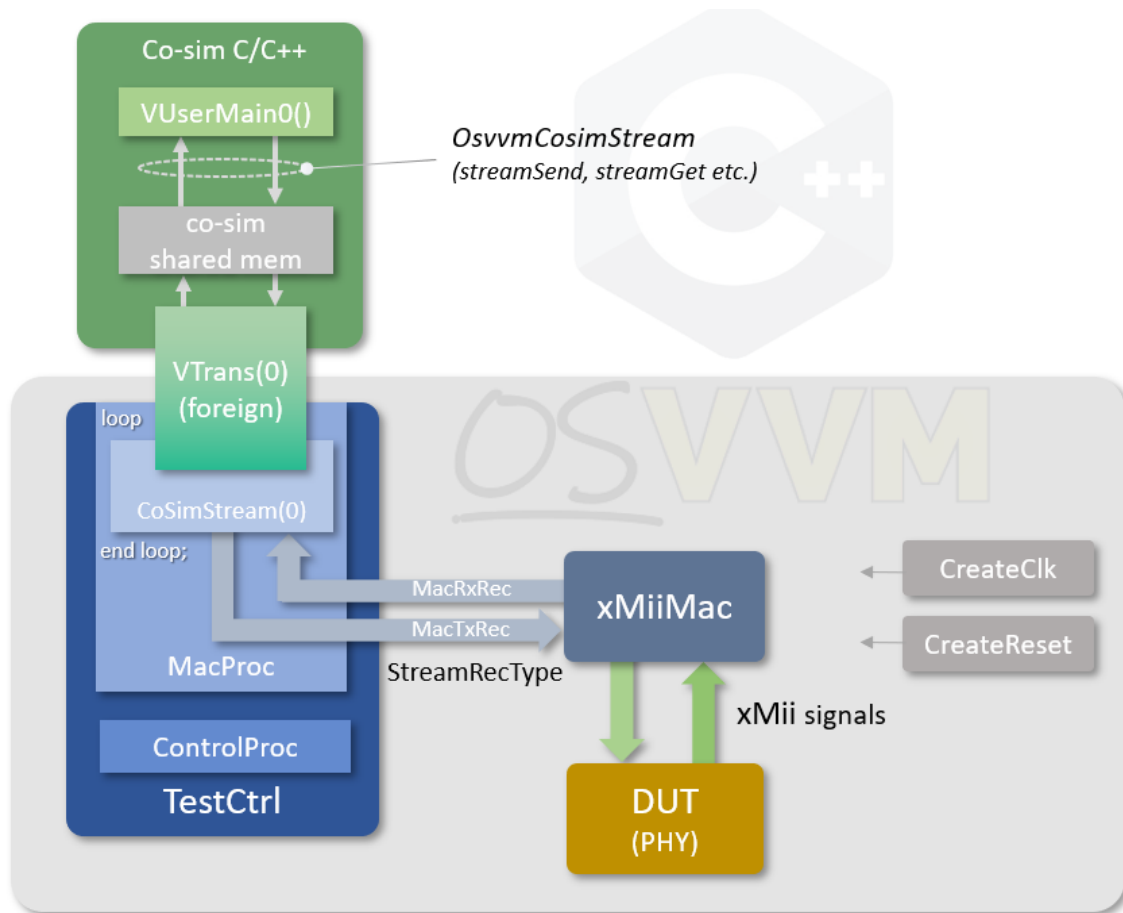


Figure 4: Co-simulation Environment Stream Example

The main features, then, of the co-simulation C++ code are as follows:

- Ability to support multiple, independent Address Bus Interfaces (such as AXI, Avalon etc.) for word and burst transfers
- Multiple nodes to drive separate AddressBusRecType transaction signals from multiple processes
- Ability to handle interrupts in the software via callbacks
- Ability to advance simulation time (without a transaction)

The rest of this document will detail how to use the co-simulation features to drive address bus transactions from the VHDL/OSVVM side and the C++ side.

2 Prerequisites for Windows

To run the co-simulation features on Windows requires the use of MSYS2 and the gcc toolchains for 64- and 32-bit compilation, as well as make and python3. The list below provides a link to the MSYS2 installer. Once installed, the required gcc modules can be installed as shown for the latest MSYS2 at time of writing (recommend updating existing installations before proceeding).

- msys2

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- mingw-w64 32- and 64-bit gcc toolchains, with required non-included libraries
- From an MSYS2 MINGW64 shell:
 - `pacman -S mingw-w64-x86_64-gcc`
 - `pacman -S mingw-w64-i686-gcc`
 - `pacman -S mingw-w64-x86_64-dlfcn`
 - `pacman -S mingw-w64-i686-dlfcn`
- Utility functions
- From an MSYS2 MINGW64 shell:
 - `pacman -S make`
 - `pacman -S python3`
 - `pacman -S rlwrap` (for GHDL and NVC)
 - `pacman -S mingw-w64-x86_64-tcllib` (for GHDL and NVC)

Note that **ModelSim** require running from the MSYS2 MINGW32 shell, as it is a 32-bit executable and the co-simulation code must be compiled for 32-bits, whereas all the other simulators require using the MSYS32 MINGW64 shell for 64-bit compilation.

If **GHDL** is required to be used, the `mingw-w64-x86_64-ghdl-llvm` package must be installed with `pacman` and should be run from the MSYS2 MINGW64 shell. It must also be ensured that `tclsh` is available (`pacman -S tcl`).

For **Active-HDL** or **Riviera-PRO** to use the local MSYS2 environment and toolchain the tool installation's `mingw` folder should be renamed (e.g., to `_mingw`) to prevent interference. Alternatively `PATH` can be setup to point to `<path to Aldec tool>/mingw/bin` to use the bundled toolchain and the `LD_LIBRARY_PATH` updated with `<path to Aldec tool>/mingw/x86_64-w64-mingw32/lib`.

3 Running the Co-simulation Examples

See the steps in the OSVVM Overview guide for running the OSVVM demos. Do the steps shown in Figure 5 in your simulator. `StartUp.tcl` needs to be sourced each time you start the simulator. See the Script User Guide for additional details for Aldec's ActiveHDL and GHDL.

```
cd sim
source ../OsvvmLibraries/Scripts/StartUp.tcl
build ../OsvvmLibraries/OsvvmLibraries.pro
build ../OsvvmLibraries/CoSim/RunAllTests.pro
```

Figure 5. Compiling and Running Co-simulation tests

4 Co-simulation Supported Platforms

Below is a list of simulators and the tested support for them for the co-simulation features of OSVVM.

Vendor	Simulator	Linux	Windows
Siemens	ModelSim	✓	✓
Siemens	QuestaSim	✓	✓
Open Source	GHDL	✓	✓
Aldec	Active-HDL	N/A	✓
Aldec	Riviera-PRO	✓	✓
Open Source	NVC	✓	✓

Table 1: Supported Simulators

5 OSVVM Co-simulation Address Transaction Procedure

To add address bus transaction co-simulation to the VHDL test logic in OSVVM, procedures are provided in `OsvvmTestCoSimPkg`. The procedures stand in for address bus transaction generation logic such as that demonstrated, for example, in the `AXI4/Axi4/TestCases/TbAxi4_MemoryReadWrite.vhd` test. The co-simulation procedures use the standard OSVVM test environments driving the `AddressBusRecType` records, usually from a manager process, to instigate bus reads and writes on, for instance, AXI4 or Axi4Lite busses, as dictated by the test bench used. The instigators of the transactions are then from C++ programs compiled and run with the simulation instead of direct calls to the OSVVM Address Bus Transaction procedures [1]. The co-simulation demonstration programs use the `TbAxi4Memory.vhd` test benches for Axi4 and Axi4Lite.

Before any calls to co-simulated code can be made, the code must be initialised for *all* nodes to be used. Separate programs can be run concurrently if needed (though not usual), and each must have a unique node ID. This is done using the `CoSimInit` procedure, as shown below.

```

-----
procedure CoSimInit (
-----
    variable NodeNum          : in    integer := 0
) ;

```

The procedure would normally be called from within the `ControlProc` of the test case, or possibly from within the `ManagerProc` if only a single node used, before any calls made to other co-simulation procedures.

To generate address bus transactions from the co-simulation software repeated calls to a procedure `CoSimTrans` must be made from a process (in a loop for example) that drive an `AddressBusRecType` signal. This would normally be from a test case's `ManagerProc`, but multiple driving processes are allowed and each can make calls to `CoSimTrans`, so long as each uses a unique node ID and that node has been initialised with a call to `CoSimInit` (see, for example,

CoSim/testbench/TestCases_Interrupt/TbAb_InterruptCosim1.vhd). The CoSimTrans procedure is defined below:

```
-----  
procedure CoSimTrans (  
-----  
    signal  ManagerRec      : inout  AddressBusRecType ;  
    variable Done           : inout  integer ;  
    variable Error          : inout  integer ;  
    variable IntReq         : in      integer := 0 ;  
    variable NodeNum        : in      integer := 0  
) ;
```

The main argument to the procedure is the ManagerRec that is used to send a transaction request to the address bus transaction VC, such as an AXI4 manager VC. The Done, and Error arguments indicate status from the co-simulation software. These must be connected to local variables in the calling process in order for state to be preserved between calls. The calling procedure can choose to use these signals or not, but that indicate state that is useful in controlling the simulation from state that the software has indicated.

The Done argument is an indication that the co-simulation software has completed its run, when non-zero, and will no longer generate output. The calling process may use this to halt testing, but the status is an indication that the software is complete, rather than a request to halt. In systems with multiple nodes, a test may wish only to stop when all nodes have indicated that they are done.

The co-simulation software can also have internal errors and self-check failures. It communicates this by setting the Error argument to a non-zero value. The calling process can use this to assert a failure and even stop at the first instance this error state.

The IntReq input argument is used to communicate interrupt status to the co-simulation software. This boolean, when true, will call a function in the co-simulation software if one has been registered as an interrupt callback function (see the next section). The callback function is called for every call to CoSimTrans that the input is true, so if the source of the interrupt in the test bench is a level driven interrupt, the calling process should only set the IntReq true when the state changes and the callback to set/clear interrupt state to the software on alternate calls. An edge driven interrupt would need the IntReq true just for cycles on its active edge. For environments that do not use the interrupt callback features, the argument can be left off and the default false value used.

The NodeNum argument is the node ID. When only one co-simulation process present, this can be left off and the node defaults to 0. Where multiple processes are generating transactions each must use a unique NodeNum.

This, then, is all that is required from the VHDL/OSVVM environment side to be able to generate transactions for both word/sub-word and burst address bus transactions.

6 OSVVM Address Bus Transaction C++ API

The user side C++ code has access to an API in order to generate address bus transactions, allow simulation time to advance and to generate status. It can also get interrupt status from the simulation if present. A simple example for a program is given in Figure 7. For each node used in the OSVVM test bench, there is expected to be a user written function named `VUserMain $n$` , where n is the number matching the node ID used (e.g. `VUserMain0`). This is like the `main()` of a C/C++ program for the node, or perhaps more like `WinMain()` for Windows non-console programs. An example definition is shown below:

```
-----
extern "C" void VUserMain0()
-----
{
// User code here
}
```

The source code for the user code would normally be gathered into a single sub-directory for a given test. This might include just a single source file (e.g. `VUserMain0.cpp`) but can have sub-files to any complexity and would also include the code for all active nodes. All the source code then has access to the co-simulation API.

6.1 OsvvmCosim Class Constructor

The API is defined in the `OsvvmCosim.h` header in `CoSim/code`. This is a C++ wrapper class to abstract away the lower level calls to the co-simulation infrastructure software. A constructor defines which node the object is attached to.

```
-----
OsvvmCosim (
-----
    int nodeIn = 0,
    std::string test_name = ""
);
```

Any user file can construct the API object. This class is a wrapper and does not hold state (except the node number) and so sub-programs can create their own API objects (with the correct node for the top level program) to gain access to the API methods.

In order to identify the different co-simulation tests, that may all uses the same underlying test bench, the software can set a test name when constructing an `OsvvmCosim` object using the `test_name` argument. By default, the argument is set to "" and the test name will not be set. Any other value will be used to set the test name in the OSVVM environment.

6.2 Advancing Time

Advancing time without generating transactions is done with the `tick()` method as defined below.

```
-----  
int tick (  
-----  
    const int ticks,  
    const bool done = false,  
    const bool error = false  
);
```

The method is called with a ticks parameter to indicate how many calls to the CoSimTrans VHDL procedure should be made without generating a transaction. If the calling process does not add any additional clock waits in the loop calling CoSimTrans, then the ticks will be clock cycles. The optional done and error inputs send status to the equivalent arguments on CoSimTrans as explained in the previous section.

The transaction methods are the methods that are used to generate the address bus activity.

6.3 Transaction methods

The basic available write and read transaction methods are listed below for both single word transactions and burst transactions.

```
-----  
uint<nn>_t transWrite (  
-----  
    const uint<nn>_t addr,  
    const uint<nn>_t data,  
    const int        prot = 0 );
```

```
-----  
uint<nn>_t transWriteAsync (  
-----  
    const uint<nn>_t addr,  
    const uint<nn>_t data,  
    const int        prot = 0 );
```

```
-----  
uint<nn>_t transWriteAndRead (  
-----  
    const uint<nn>_t addr,  
    const uint<nn>_t wdata,  
    const uint<nn>_t *rdata,  
    const int        prot = 0 );
```

```
-----
uint<nn>_t transWriteAndReadAsync (
-----
```

```
    const uint<nn>_t addr,
    const uint<nn>_t wdata,
    const int        prot = 0 );
-----
```

```
void transRead (
-----
```

```
    const uint<nn>_t addr,
    const uint<nn>_t *data,
    const int        prot = 0);
-----
```

```
bool transReadPoll (
-----
```

```
    const uint<nn>_t addr,
    const uint<nn>_t *data,
    const int        idx,
    const int        bitval,
    const int        waittime = 10,
    const int        prot = 0
);
-----
```

```
void transBurstWrite (
-----
```

```
    const uint<nn>_t addr,
    const int        bytesize,
    const int        prot = 0);
-----
```

```
void transBurstWriteAsync (
-----
```

```
    const uint<nn>_t addr,
        uint8_t      *data,
    const int        bytesize,
    const int        prot = 0);
-----
```

```
void transBurstRead (
-----
```

```
    const uint<nn>_t addr,
        uint8_t      *data,
    const int        bytesize,
    const int        prot = 0);
-----
```

These transaction methods are overloaded for various sizes of address and data , using the `stdint.h` definitions, where `uint<nn>_t` indicates different uint sizes. For address parameters this is `uint32_t` or

uint64_t. For the data parameters of transRead and transWrite this is one of uint8_t, uint16_t, uint32_t or uint64_t (if address type is also uint64_t). Note that using uint64_t can only be done if the target test bench is configured for a 64-bit bus architecture.

The single word methods, transRead and transWrite, and its non-blocking asynchronous equivalent transWriteAsync, take an address, data and optional prot (protection) arguments, with the transRead having a pointer to the variable for returning the read data. The transWrite returns a value which is a future-proofing for returning read data in the same transaction as a write, but is not yet supported. The transRead method does not return a value and a pointer is used to return the data as this is how the method selects the size of the read transaction (byte, half-word etc.) based on the argument's size. The transWriteAndRead method does a write to the specified address with the wdata value, and returns the old value in the rdata argument. The transWriteAndReadAsync method is similar, but the read data is not returned at the call. The read transaction is instigated, however, and the data can be fetched with methods such as transReadData. The transReadPoll method will poll a memory location specified by the addr argument, at intervals defined by the waittime argument, until the data read has its bit indexed at idx matching the value in bitval. The value read on matching is returned in data.

The burst transactions use byte buffers to send and return data. The calling code provides buffer space to a maximum of 4Kbytes preloaded with byte data for writes, or updated with byte data on reads. If no data is specified, then the burst write will use whatever has been pushed to the write fifo using transBurstPushData (see later). The bytesize parameter indicates the transfer size. The co-simulation code will not go beyond 4Kbytes when reading or writing to the buffers, so it is recommended to use 4Kbyte buffers in all cases to avoid overrun. A non-blocking version of transBurstWrite is provided as transBurstWriteAsync. In addition, the burst writes can have predetermined patterns as a source of data, namely incrementing or random, with a data byte provided to indicate the first word to use for the pattern. The methods are shown below:

```
-----  
void transBurstWriteIncrement (
```

```
-----  
    const uint<nn>_t addr,  
    const uint8_t    firstbyte,  
    const int        bytesize,  
    const int        prot = 0);
```

```
-----  
void transBurstWriteIncrementAsync (
```

```
-----  
    const uint<nn>_t addr,  
    const uint8_t    firstbyte,  
    const int        bytesize,  
    const int        prot = 0);
```

```
-----
void transBurstWriteRandom (
-----
    const uint<nn>_t addr,
    const uint8_t    firstbyte,
    const int        bytesize,
    const int        prot = 0);
```

```
-----
void transBurstWriteRandomAsync (
-----
    const uint<nn>_t addr,
    const uint8_t    firstbyte,
    const int        bytesize,
    const int        prot = 0);
```

6.3.1 Split Transactions

For VCs that support it, methods are provided for split transactions—i.e., sending address and data separately, at different times. The methods to support this are listed below:

```
-----
void transWriteAddressAsync (
-----
    const uint<nn>_t addr,
    const int        prot = 0);
```

```
-----
uint<nn>_t transWriteDataAsync (
-----
    const uint<nn>_t data,
    const uint32_t    bytelane,
    const int        prot = 0 );
```

```
-----
void transReadAddressAsync (
-----
    const uint<nn>_t addr,
    const int        prot = 0);
```

```
-----
void transReadData (
-----
    const uint<nn>_t *data
);
```

```
-----  
bool transTryReadData (  
-----  
    const uint<nn>_t *data  
);
```

The transTryReadData is the same as transReadData, but will return if no data available, rather than block. It will return false if no data available and true if data was available, which is then returned in the data buffer.

6.3.2 Push and Pop Burst Data

Data can be pushed and popped from the transaction fifos independently of transaction method calls for use in split transactions etc. In addition, the fifos can be filled with predetermined patterns. The methods for this are listed below.

```
-----  
void transBurstPushData (  
-----  
    const uint8_t *data,  
    const int      bytesize);
```

```
-----  
void transBurstPushIncrement (  
-----  
    const uint8_t data,  
    const int      bytesize);
```

```
-----  
void transBurstPushRandom (  
-----  
    const uint8_t data,  
    const int      bytesize);
```

```
-----  
void transBurstPopData (  
-----  
    const uint8_t *data,  
    const int      bytesize);
```

6.3.3 Read Checks

Methods are provided that do check on data as they are read for either words or bursts. This can be using provided data, using data already pushed to the Fifo, or predetermined patterns. The provided methods for word transactions are listed below:


```
-----
void transReadCheck (
-----
    const uint<nn>_t  addr,
    const uint<nn>_t  data,
    const int        prot = 0);
-----
```

```
-----
void transReadDataCheck (
-----
    const uint<nn>_t  data);
-----
```

```
-----
bool transTryReadDataCheck (
-----
    const uint<nn>_t  data);
-----
```

The first method does a blocking read transaction, and compares the read value with that supplied as the data argument. The second method does the same check, but with already received data from a previous read. The last method is the same as the second but is non-blocking, returning false if no word available and true if data was available and was checked.

The burst transactions also have check methods, with methods that instigate a transaction and check, and methods that check previously received data. In addition, the data can be checked against a supplied set of data, or against a predetermined pattern. The provided methods are shown below:

```
-----
void transBurstReadCheckData (
-----
    const uint<nn>_t  addr,
    const uint8_t     *data,
    const int         bytesize,
    const int         prot = 0);
-----
```

```
-----
void transBurstCheckData (
-----
    const uint8_t     *data,
    const int         bytesize,
    const int         prot = 0);
-----
```

```
-----
void transBurstReadCheckIncrement (
-----
    const uint<nn>_t  addr,
    const uint8_t     data,
    const int         bytesize,
    const int         prot = 0);
-----
```

```
-----  
void transBurstReadCheckRandom (  
-----  
    const uint<nn>_t  addr,  
    const uint8_t     data,  
    const int         bytesize,  
    const int         prot = 0);
```

6.4 Directive Methods

Some directive methods are provided for transaction statistics and synchronization, as listed below:

```
-----  
int transGetTransactionCount (  
-----  
    void  
);
```

```
-----  
int transGetWriteTransactionCount (  
-----  
    void  
);
```

```
-----  
int transGetReadTransactionCount (  
-----  
    void  
);
```

```
-----  
void transWaitForTransaction (  
-----  
    void  
);
```

```
-----  
void transWaitForWriteTransaction (  
-----  
    void  
);
```

```
-----  
void transWaitForReadTransaction (  
-----  
    void  
);
```

6.5 Interrupt Callback

User code can register a callback function which will be called by the simulation every time the IntReq input to the CoSimTrans procedure changes state in the OSVVM test process. To register an interrupt callback function the following method is used:

```
-----
void regInterruptCB (
-----
    pVUserInt_t func
);
```

6.6 Additional Methods

As well as the methods outlined above a couple of other methods are provided to aid usage of the class. These are shown below:

```
-----
int getNodeNumber (
-----
    void
);

-----
int waitForSim (
-----
    void
);
```

The getNodeNumber method simply returns the node number associated with the instantiation object of the class. As stated before, multiple objects are allowed for access to a given node. To aid in managing which node a particular object has been associated with, this method gives access to that state.

The waitForSim method is for running in GHDL when in callable simulation mode. For all other modes, this call does nothing. When active, a call to this method blocks until the simulation is in a suitable state to have transaction methods called to instigate traffic on the bus. For more details see section 12.6, which also has example code using this method.

6.7 OsvvmCosimInt Interrupt Support Class

The OsvvmCosimInt class, derived from the OsvvmClass, adds additional programming support for handling interrupts for new test code that has no other means of processing interrupts such as when using a software processor model. This inherits all the methods of the parent class, as described in the sections above, including the regInterruptCB() method.

The type of callback function, as defined by pVUserInt_t, is int <myfunc>(int). When the callback is executed the int argument contains the current state of the IntReq input to the CoSimTrans procedure. Example code can be found in OsvvmLibraries/CoSim/test/interruptCB. In addition it provides methods for registering up to 32 interrupt service routine (ISR) functions:

```
-----  
void registerIsr (  
-----
```

```
    const pVUserInt_t func,  
    const int         level);
```

Each function registered to a valid level (from 0 to 31) will be called if the interrupt level indicates that it is active, it is enabled and no higher priority is active. The priority has 0 as the highest and 31 as the lowest. Various methods are provided for controlling whether an interrupt is enabled or not. A master interrupt can be enabled or disabled to globally control all interrupts.

```
-----  
void enableMasterInterrupt (  
-----
```

```
    void  
);
```

```
-----  
void disableMasterInterrupt (  
-----
```

```
    void  
);
```

Each individual level also has an enable and disable method for individual control of each ISR:

```
-----  
void enableIsr (  
-----
```

```
    const int level  
);
```

```
-----  
void disableIsr (  
-----
```

```
    const int level  
);
```

In order to update interrupt state, a method is provided to pass in the current interrupt request status:

```
-----  
void updateIntReq (  
-----
```

```
    const int intReq  
);
```

This method would normally be used in conjunction with a callback function registered with the `regInterruptCB()` method, which is called whenever the interrupt request input to `CoSimTrans` changes. This function would have access to the `OsvvmCosimInt` object and call its `updateIntReq()` method to pass in the interrupt request status.

7 OSVVM Co-simulation Address Bus Responder Procedure

As a complimentary feature to the address bus model independent transaction features detailed in the last two sections an address bus transaction responder is also provided. This is used, for example, when testing IP with a manager interface and the test environment is required to generate the responses to read and write requests. From the VHDL side, the responder re-uses the CoSimInit procedure (see section 5) to initialise a node for a responder co-simulation user software. The supplied node number must be unique for each instance that will be used, and from any other co-simulation elements for transactions and streams).

To generate address bus transactions responder from the co-simulation software repeated calls to a procedure CoSimResp must be made from a process (in a loop for example) that drive an AddressBusRecType signal. This would normally be from a test case's SubordinateProc, but multiple driving processes are allowed and each can make calls to CoSimResp, so long as each uses a unique mode ID and that node has been initialised with a call to CoSimInit (see, for example, CoSim/TestCases/TbAb_Responder.vhd). The CoSimResp procedure is defined below:

```
-----
procedure CoSimResp (
-----
    signal    SubordinateRec    : inout  AddressBusRecType ;
    variable Done               : inout  integer ;
    variable Error              : inout  integer ;
    variable NodeNum           : in     integer := 1
);
```

The main argument to the procedure is the SubordinateRec that is used to send transaction responses to the address bus transaction VC, such as an AXI4 manager VC, or to an AXI4 manager interface on a DUT. The Done, and Error arguments indicate status from the co-simulation software. These must be connected to local variables in the calling process in order for state to be preserved between calls. The calling procedure can choose to use these signals or not, but they indicate state that is useful in controlling the simulation from that indicated from software.

The Done argument is an indication that the co-simulation software has completed its run, when non-zero, and will no longer generate output. The calling process may use this to halt testing, but the status is an indication that the software is complete, rather than a request to halt. In systems with multiple nodes, a test may wish only to stop when all nodes have indicated that they are done.

The co-simulation software can also have internal errors and self-check failures. It communicates this by setting the Error argument to a non-zero value. The calling process can use this to assert a failure and even stop at the first instance this error state.

The NodeNum argument is the node ID. When only one co-simulation process present, this can be left off and the node defaults to 1 for the responder. Where multiple processes are generating transactions each must use a unique NodeNum.

8 OSVVM Address Bus Responder C++ API

The user side C++ code has access to an API in order to generate address bus responses, allow simulation time to advance and to generate status. For each node used in the OSVVM test bench, there is expected to be a user written function named `VUserMain $n$` , where n is the number matching the node ID used (e.g. `VUserMain1`, with a node number of 1 being the default for a responder). This is like the `main()` of a C/C++ program for the node, or perhaps more like `WinMain()` for Windows non-console programs. An example definition is shown below:

```
-----
extern "C" void VUserMain1()
-----
{
// User code here
}
```

The source code for the user code would normally be gathered into a single sub-directory for a given test. This might include just a single source file (e.g. `VUserMain1.cpp`) but can have sub-files to any complexity and would also include the code for all active nodes. All the source code then has access to the co-simulation API.

8.1 OsvvmCosim Class Constructor

The API is defined in the `OsvvmCosimResp.h` header in `CoSim/code`. This is a C++ wrapper class to abstract away the lower level calls to the co-simulation infrastructure software. A constructor defines which node the object is attached to.

```
-----
OsvvmCosimResp (
-----
    int nodeIn = 0,
    std::string test_name = ""
);
```

Any user file can construct the API object. This class is a wrapper and does not hold state (except the node number) and so sub-programs can create their own API objects (with the correct node for the top level program) to gain access to the API methods.

In order to identify the different co-simulation tests, that may all uses the same underlying test bench, the software can set a test name when constructing an `OsvvmCosim` object using the `test_name` argument. By default, the argument is set to "" and the test name will not be set. Any other value will be used to set the test name in the OSVVM environment.

8.2 Advancing Time

Advancing time without generating transactions is done with the `tick()` method as defined below.

```
-----
int tick (
-----
    const int ticks,
    const bool done = false,
    const bool error = false
);
```

The method is called with a ticks parameter to indicate how many calls to the CoSimTrans VHDL procedure should be made without generating a transaction. If the calling process does not add any additional clock waits in the loop calling CoSimTrans, then the ticks will be clock cycles. The optional done and error inputs send status to the equivalent arguments on CoSimTrans as explained in the previous section.

8.3 Response methods

The response methods are the methods that are used to generate the address bus response activity. The basic available write and read transaction methods are listed below:

```
-----
void respGetWrite (
-----
    uint<nn>_t *addr,
    uint<nn>_t *data
);
```

```
-----
bool respTryGetWrite (
-----
    uint<nn>_t *addr,
    uint<nn>_t *data
);
```

```
-----
void respSendRead (
-----
    uint<nn>_t *addr,
    uint<nn>_t data
);
```

```
-----
bool respTrySendRead (
-----
    uint<nn>_t *addr,
    uint<nn>_t data
);
```

These response methods are overloaded for various sizes of address and data, using the `stdint.h` definitions, where `uint<nn>_t` indicates different uint sizes. For address parameters this is `uint32_t` or `uint64_t`. For the data parameters of `transRead` and `transWrite` this is one of `uint8_t`, `uint16_t`, `uint32_t` or `uint64_t` (if address type is also `uint64_t`). Note that using `uint64_t` can only be done if the target test bench is configured for a 64-bit bus architecture.

The methods `respGetWrite` and `respSendRead`, and their non-blocking asynchronous equivalents `respTryGetWrite`, and `respTrySendData`, take an address pointer argument, which returns the address associated with the received transaction. The write methods take a data pointer argument in which the written data is returned. The read methods take a data input argument to send as the read response data. The `respGetWrite` and `respSendRead` methods are blocking, and will not return until the transaction is completed. For writes, this means that a whole write transaction has been received, and for reads, that the response data has been sent. The try methods return a boolean status. If the response can't be processed it will return false. For writes, this means no write data is available. For reads, this means that no read address has been received.

8.3.1 Split Response Methods

For VCs that support it, methods are provided for split transactions—i.e., fetching (or sending, where applicable) address and data separately, at different times. The methods to support this are listed below.

```
-----
void respGetWriteAddress (
-----
    uint<nn>_t *addr
);

-----

void respTryGetWriteAddress (
-----
    uint<nn>_t *addr
);

-----

void respGetWriteData (
-----
    uint<nn>_t *data
);

-----

void respTryGetWriteData (
-----
    uint<nn>_t *data
);
```



```
-----
void respGetReadAddress (
-----
    uint<nn>_t *addr
);
```

```
-----
void respTryGetReadAddress (
-----
    uint<nn>_t *addr
);
```

```
-----
void respSendReadData (
-----
    uint<nn>_t data
);
```

```
-----
bool respSendReadDataAsync (
-----
    uint<nn>_t data
);
```

There are methods for fetching both read and write received addresses, with `respGetWriteAddress` and `respGetReadAddress` being block, waiting until an address is available. The non-blocking versions (`respTryGetWriteAddress` and `respTryGetReadAddress`) return false if no address is available. Retrieving write data is done with `respGetWriteData` (blocking) and `respTryGetWriteData` (non-blocking, returning false if no write data available). To send the response data, `respSendReadData` (blocking until a read address is available) and `respSendReadData` (non-blocking). This last does return the status of whether a read address was available at the point the methods was called, but the data is placed in the respond fifo regardless.

8.4 Directive Methods

Some directive methods are provided for transaction statistics and synchronization, as listed below:

```
-----
int respGetTransactionCount (
-----
    void
);
```

```
-----  
int respGetWriteTransactionCount (  
-----  
    void  
);
```

```
-----  
int respGetReadTransactionCount (  
-----  
    void  
);
```

```
-----  
void respWaitForTransaction (  
-----  
    void  
);
```

```
-----  
void respWaitForWriteTransaction (  
-----  
    void  
);
```

```
-----  
void respWaitForReadTransaction (  
-----  
    void  
);
```

8.5 Additional Methods

As well as the methods outlined above a couple of other methods are provided to aid usage of the class. These are shown below:

```
-----  
int getNodeNumber (  
-----  
    void  
);
```

```
-----  
int waitForSim (  
-----  
    void  
);
```

The `getNodeNumber` method simply returns the node number associated with the instantiation object of the class. As stated before, multiple objects are allowed for access to a given node. To aid in managing which node a particular object has been associated with, this method gives access to that state.

The `waitForSim` method is for running in GHDL when in callable simulation mode. For all other modes, this call does nothing. When active, a call to this method blocks until the simulation is in a suitable state to have transaction methods called to instigate traffic on the bus. For more details see section 12.6, which also has example code using this method.

9 OSVVM Co-simulation Stream Transaction Procedure

To enable streaming transaction co-simulation in the VHDL test logic of OSVVM, procedures are provided in `OsvvmTestCoSimPkg`. The procedures stand in for streaming transaction generation logic such as that demonstrated, for example, in the `Ethernet/TestStandAlone/Tb_xMii.vhd` test. The co-simulation procedures use the standard OSVVM test environments driving the `StreamRecType` records, usually from a manager process, to instigate bus reads and writes on, for instance, Ethernet or UART interconnects, as dictated by the test bench used. The instigators of the transactions are then from C++ programs compiled and run with the simulation instead of direct calls to the OSVVM Streaming Transaction procedures [3].

Just as for the address bus transactions, before any calls to co-simulated code can be made, the code must be initialised for *all* nodes to be used. Separate programs can be run concurrently if needed (though not usual), and each must have a unique node ID. This is done using the `CoSimInit` procedure, as shown below.

```
-----  
procedure CoSimInit (  
-----  
    variable NodeNum          : in    integer := 0  
) ;
```

To generate address bus transactions from the co-simulation software repeated calls to a procedure `CoSimStream` must be made from a process (in a loop for example) that drive a `StreamRecType` signal. This would normally be from a test case's `ManagerProc`, but multiple driving processes are allowed and each can make calls to `CoSimStream`, so long as each uses a unique model ID and that node has been initialised with a call to `CoSimInit`. The `CoSimStream` procedure is defined below:

```
-----  
procedure CoSimStream (  
-----  
    signal  TxRec          : inout StreamRecType ;  
    signal  RxRec          : inout StreamRecType ;  
    variable Done          : inout integer ;  
    variable Error         : inout integer ;  
    variable NodeNum       : in    integer := 0) ;
```

The main arguments to the procedure are the `TxRec` and `RxRec` that are used to send transaction requests and receive transactions to and from a streaming transaction VC, such as an Ethernet VC. The `Done`, and `Error` arguments indicate status from the co-simulation software. These must be connected

to local variables in the calling process in order for state to be preserved between calls. The calling procedure can choose to use these signals or not, but that indicate state that is useful in controlling the simulation from state that the software has indicated.

The Done argument is an indication that the co-simulation software has completed its run, when non-zero, and will no longer generate output. The calling process may use this to halt testing, but the status is an indication that the software is complete, rather than a request to halt. In systems with multiple nodes, a test may wish only to stop when all nodes have indicated that they are done.

10 OSVVM Streaming C++ API

Like for the Address Bus Transaction, the streaming features have a C++ API in order to generate streaming transactions, allow simulation time to advance and to generate status. User code is also entered via a `VUserMainn` function, with n being the node number (defaulting to 0).

10.1 OsvvmCosimStream Class Constructor

The Streaming API is provided via a class, `OsvvmCosimStream`, defined in `OsvvmCosimStream.h`, which can be found in `CoSim/code`. This is a C++ wrapper class to abstract away the lower level calls to the co-simulation infrastructure software. A constructor defines which node the object is attached to.

```
-----  
OsvvmCosimStream (  
-----  
    int nodeIn = 0,  
    std::string test_name = ""  
);
```

Any user file can construct the API object. This class is a wrapper and does not hold state (except the node number) and so sub-programs can create their own API objects (with the correct node for the top level program) to gain access to the API methods. In order to identify the different co-simulation tests, that may all uses the same underlying test bench, the software can set a test name when constructing an `OsvvmCosim` object using the `test_name` argument. By default, the argument is set to "" and the test name will not be set. Any other value will be used to set the test name in the OSVVM environment.

10.2 Advancing Time

Advancing time without generating transactions is done with the `tick()` method as defined below.

```
-----  
int tick (  
-----  
    const int ticks,  
    const bool done = false,  
    const bool error = false );
```

The method is called with a `ticks` parameter to indicate how many calls to the `CoSimStream` VHDL procedure should be made without generating a transaction. If the calling process does not add any

additional clock waits in the loop calling CoSimStream, then the ticks will be clock cycles. The optional done and error inputs send status to the equivalent arguments on CoSimStream as explained in the previous section. The transaction methods are the methods that are used to generate the streaming activity.

10.3 Transaction Methods

The basic streaming transaction methods are shown below:

```
-----  
uint<nn>_t streamSend (  
-----  
    const uint<nn>_t data,  
    const int        param = 1  
);
```

```
-----  
uint<nn>_t streamSendAsync (  
-----  
    const uint<nn>_t data,  
    const int        param = 1  
);
```

```
-----  
void streamGet (  
-----  
    uint<nn>_t *data  
);
```

```
-----  
void streamGet (  
-----  
    uint<nn>_t *data,  
    const int  *status  
);
```

```
-----  
void streamBurstSend (  
-----  
    uint8_t    *data,  
    const int   bytesize  
);
```

```
-----  
void streamBurstSend (  
-----  
    const int   bytesize  
);
```

```
-----
void streamBurstSendAsync (
-----
    uint8_t    *data,
    const int   bytesize
);
```

```
-----
void streamBurstSendAsync (
-----
    const int   bytesize
);
```

```
-----
void streamBurstGet (
-----
    uint8_t    *data,
    const int   bytesize
);
```

The streamSend and streamGet transaction methods are overloaded for various sizes of data , using the stdint.h definitions, where uint<nn>_t indicates different uint sizes. For the data parameter, this is one of uint8_t, uint16_t, uint32_t or uint64_t. There are also non-bloking variants of the of the transSend and transBurstSend methods; transSendAsync and transBurstSendAsync. In addition the transBurstSend and transBurstSendAsync can have a data buffer pointer argument or not. When supplied the buffer data is pushed to the transaction fifo before the transaction is instigated. When no data buffer argument is given, the data already in the fifo is used, loaded previously via the push methods (see below).

The streamSend and streamSendAsync optional param argument is used by some VCs. For example, the UART VC uses this to inject parity errors and other conditions. The default value is for no errors. The streamGet also has an optoinal second argument, status, which can be used to get status returned from a streaming VC, if it supports this. For example, the UART VC can indicate receiving a a transaction with a parity error.

There are also pattern generating versions of the burst send methods. These can generate patterns in the fifo as either incrementing or random, with a data argument specifying the first byte of the pattern. The methods are listed below.

```
-----
void streamBurstSendIncrement (
-----
    uint8_t    data,
    const int   bytesize,
    const int   param = 1
) ;
```

```
-----
void streamBurstSendIncrementAsync (
-----
```

```
    uint8_t  data,
    const int  bytesize,
    const int  param = 1
) ;
-----
```

```
void streamBurstSendRandom (
-----
```

```
    uint8_t  data,
    const int  bytesize,
    const int  param = 1
) ;
-----
```

```
void streamBurstSendRandomAsync (
-----
```

```
    uint8_t  data,
    const int  bytesize,
    const int  param = 1
) ;
-----
```

The streaming burst transactions use byte buffers to send and return data. The calling code provides buffer space to a maximum of 4Kbytes preloaded with byte data for writes, or updated with byte data on reads. The bytesize parameter indicates the transfer size. The co-simulation code will not go beyond 4Kbytes when reading or writing to the buffers, so it is recommended to use 4Kbyte buffers in all cases to avoid overrun.

10.3.1 Push and Pop Methods

Various methods are provided to push and pop data from the transaction fifos, both Tx and Rx. The push methods are either for transmission and are pushed to the Tx fifo, or are for comparison with received data, when they push to the Rx fifo (these are the push to check methods). There are also methods to load the fifos with predetermined patterns, namely increment and random patterns, with the first byte specified as a parameter. The methods are listed below:

```
-----
void streamBurstPushData (
-----
```

```
    uint8_t *data,
    const int  bytesize
) ;
-----
```

```
void streamBurstPushCheckData (
-----
```

```
    uint8_t *data,
    const int  bytesize
) ;
-----
```

```

-----
void streamBurstPushIncrement (
-----
    uint8_t data,
    const int    bytesize
) ;
-----

void streamBurstPushCheckIncrement (
-----
    uint8_t data,
    const int    bytesize
) ;
-----

void streamBurstPushRandom (
-----
    uint8_t data,
    const int    bytesize
) ;
-----

void streamBurstPushCheckRandom (
-----
    uint8_t data,
    const int    bytesize
) ;
-----

void streamBurstPopData (
-----
    uint8_t *data,
    const int    bytesize
) ;

```

10.3.2 Check Methods

A set of methods are provided to check data word and burst data in the receive fifo, and these are listed below.

```

-----
void streamCheck (
-----
    const uint<nn>_t data,
    const int    param = 0
) ;

```



```
-----
void streamBurstCheck (
-----
```

```
    uint8_t  *data
    const int    bytesize,
    const int    param = 1
) ;
```

```
-----
void streamBurstCheck (
-----
```

```
    const int    bytesize,
    const int    param = 1
) ;
```

```
-----
void streamBurstCheckIncrement (
-----
```

```
    uint8_t  data
    const int    bytesize,
    const int    param = 1
) ;
```

```
-----
void streamBurstCheckRandom (
-----
```

```
    uint8_t  data
    const int    bytesize,
    const int    param = 1
) ;
```

10.3.3 Non-blocking Get and Check

Methods are provided which are non-blocking get and check operations. These return false if no data is available, otherwise they return true and return or check the data. These are listed below.

```
-----
bool streamTryGet (
-----
```

```
    uint<nn>_t *data
) ;
```

```
-----
bool streamTryGet (
-----
```

```
    uint<nn>_t *data,
    int        *status
) ;
```

```
-----  
bool streamTryCheck (  
-----  
    uint<nn>_t *data,  
    const int   param = 0  
) ;  
-----  
bool streamBurstTryGet (  
-----  
    uint<nn>_t *data,  
    const int   bytesize,  
    const int   param = 1  
) ;  
-----  
bool streamBurstTryGet (  
-----  
    const int   bytesize,  
    const int   param = 1  
) ;  
-----  
bool streamBurstTryCheck (  
-----  
    uint<nn>_t *data,  
    const int   bytesize,  
    const int   param = 1  
) ;  
-----  
bool streamBurstTryCheck (  
-----  
    const int   bytesize,  
    const int   param = 1  
) ;  
-----  
bool streamBurstTryIncrement (  
-----  
    uint8_t     data,  
    const int   bytesize,  
    const int   param = 1  
) ;
```

```
-----  
bool streamBurstTryRandom (  
-----  
    uint8_t    data,  
    const int  bytesize,  
    const int  param = 1  
) ;
```

The class also has `getNodeNumber` and `waitForSim` methods as for the `OsvvmClass`, with the same functionality (see section 6.6).

10.3.4 Directive Methods

Some additional directive methods are provided for statistics and flow synchronisation, as listed below:

```
-----  
int streamGetRxTransactionCount (  
-----  
    void  
) ;
```

```
-----  
int streamGetTxTransactionCount (  
-----  
    void  
) ;
```

```
-----  
void streamWaitForRxTransaction (  
-----  
    void  
) ;
```

```
-----  
void streamWaitForTxTransaction (  
-----  
    void  
) ;
```

11 Compiling Co-simulation Code

11.1 Compiling Basic Programs

To compile the co-simulation files, both OSVVM and C++, some additional steps are required over a pure OSVVM environment. Some Tcl procedures are available to compile the C++ code which are gathered into `CoSim/Scripts/MakeVproc.tcl`. Ultimately, these scripts call the `makefile` in the `CoSim` directory, setting parameters as appropriate. This builds the co-simulation code into a shared object called `VProc.so`, and the user code is compiled to `VUser.so`.

For normal OSVVM test case code, this might use `RunTest` to analyse and then run the test code. For a co-simulation test additional arguments must be given to compile the C++ code. The `MkVproc` TCL function is used and has two arguments:

```
-----  
proc MkVproc {  
-----  
    testname  
    {libname ""}  
}
```

The first argument is a path to the directory where the user source code is located, containing all the user code for the particular test. An optional `libname` argument allows for libraries within `CoSim/lib` to be linked with the user code by tagging a `-l:<libname>` argument. An example `.pro` script to compile a co-simulation test is shown below, where the script is located in `AXI4/Axi4/TestCases`:

```
MkVproc $::osvvm::OsvvmCoSimDirectory/tests/usercode_size  
simulate TbAxi4_CoSim
```

Figure 6: Compiling Co-simulation Code

In this example, the test case, `TbAxi4_Cosim`, is run with arguments to build the software to run with it. The `MkVproc` call specified that the top level test directory is `CoSim` (using the global `$::osvvm::OsvvmCoSimDirectory`), and that the particular test code to be compiled is in `tests/usercode_size`, under the top level test directory. The output of the make process, instigated by `MkVproc`, will be in the directory from which the `.pro` script was called, which would normally be outside of the `OsvvmLibraries` directory. The co-simulation shared objects, `VProc.so` and `VUser.so`, along with the intermediate compiled files and static libraries are all output to the running directory. With a call to `MkVproc` a clean compile is done. That is so all the build files of any previous compilation are deleted first in order to ensure no clashes or pick up of old files in the event of a compilation error. A `MkVprocNoClean` procedure is provided to skip the cleaning step for diagnostic purposes but would not be used in normal operations.

11.2 Compiling and Linking with External Libraries

For more complex programs, requiring external library linkage, OSVVM provides a make file that can be configured to compile in external code. The example below shows a usage of the make file script that compiles in a local library, `libjfixxx.a`, where `xxx` is the suffix relevant for the target OS, and an external GTK+ library. The example is specific to the Riviera-PRO simulator, but the procedure is the same for the other supported tools:

```
# Set simulator flags for the specific simulator (e.g. Riviera-PRO)
set sim RivieraPRO
set aldecpath [file normalize [info nameofexecutable]/../..]

# Get the GTK+ C and link flags using pkg-config
set gtkcflags [ exec pkg-config.exe --cflags gtk+-3.0 ]
set gtkldflags [ exec pkg-config.exe --libs  gtk+-3.0 ]

# Point to the user program source directory
set usrcdir $::osvvm::OsvvmCoSimDirectory/tests/cosim_demo

# Set the C flags for the user program, incorporating the GTK+ C flags
set usrflags "-I $::osvvm::OsvvmCoSimDirectory/include \
    ${gtkcflags}"

# Set the linker flags for the JFIF and GTK+ libraries
set usrlflags "-L $::osvvm::OsvvmCoSimDirectory/lib \
    -ljfifwin64 ${gtkldflags}"

# Clean the test directory of previous source code artifacts
exec make -C $::osvvm::OsvvmCoSimDirectory \
    OPDIR=$::osvvm::CurrentSimulationDirectory clean

# Compile the co-simulation software and user program
exec make -C $::osvvm::OsvvmCoSimDirectory \
    OPDIR=$::osvvm::CurrentSimulationDirectory \
    SIM=${sim} \
    ALDECDIR=${aldecpath} \
    USRCDIR=${usrcdir} \
    USRFLAGS=${usrflags} \
    USRLDFLAGS=${gtkldflags}
```

The above script is more verbose than might normally be necessary to clearly show each step, with lots of individual variables set up first, but the key is just the highlighted area with the call to make, using the OSVVM co-simulation makefile. Pre-defined OSVVM variables are liberally used to help here.

The output directory (OPDIR), where the compiled shared object libraries will be placed, is specified using the `$osvvm::CurrentDirectory` variable which points to where the simulation is being run. The SIM make variable sets the specific simulator being used (Riviera-PRO in this example) and ALDECDIR, specific to the ALDEC simulator, sets the path to where VHPI libraries for the simulator are located, needed for the co-simulation code—though this was worked out using OSVVM's `nameofexecutable` procedure and the library path defined relative to this, so the simulator's installation directory does not need to be coded in the script.

Then the location of the user source code is specified with USRCDIR, using the supplied `$osvvm::OsvvmCoSimDirectory` variable. A set of compile flags is defined using USRFLAGS, if any needed for the user code. In this example these are the include path for the CoSim/include directory and the paths and flags for the GTK+ library which, for this particular library, can be generated with the `pkg-config` command. Linker flags can also be set using USRLDFLAGS and, in this case, they are the auto-generated flags for GTK+ and a path to CoSim/lib and to link an internal JFIF library for JPEG image decoding.

Using the supplied make file is a little more complicated to use than MkVproc, but it allows for more elaborate compilations whilst still hiding most of the details of compiling OSVVM co-simulation code.

12 Usage Examples

In this section will be described a set of usage cases for the co-simulation address bus features. Each has a demonstration test case which may be referred to as a reference for each of the case.

12.1 C++ test code for word and burst transactions

At its simplest, the co-simulation features allow address bus transaction tests to be written in C++. The user code provides a VUserMain0 function, instantiates an OsvvmCosim object and then has access to all the API features described above. The diagram in Figure 2 shows this test setup.

Demo tests for both word/sub-word and burst transactions exist, with the C++ source as:

- CoSim/tests/usercode_size/VUserMain0.cpp
- CoSim/tests/usercode_burst/VUserMain0.cpp

Both of these programs can be run on the TbAxiMemory.vhd test bench for both Axi4Lite and Axi4. The Axi4Lite and Axi4 test case VHDL for the tests are:

- AXI4/Axi4Lite/testbench/TbAxi4_CoSim.vhd
- AXI4/Axi4/TestCases/TbAxi4_CoSim.vhd

These VHDL test case files are common to most of the relevant tests as it is just different C++ programs that need to be run on them. A simple example of a program using the word/sub-word API calls is show below.

```

#include "OsvvmCosim.h"

extern "C" void VUserMain0() {
    bool            error = false;
    uint32_t        wdata = 0;
    OsvvmCosim      cosim(0);

    for (int loop = 0; loop < 4; loop++)
    {
        uint32_t addr = 0x10000000;
        uint32_t rdata;

        for (int idx = 0; idx < 4; idx++) {
            cosim.transWrite(addr, wdata + idx);
            addr += 4;
        }

        addr = 0x10000000;

        for (int idx = 0; idx < 4; idx++) {
            cosim.transRead(addr, &rdata);
            if (rdata != (wdata + idx)) {
                error = true;
                break;
            }
            addr += 4;
        }
        wdata += 0x10;
    }

    // Flag to the simulation we're finished, after 10 more iterations
    cosim.tick(10, true, error);

    // If ever got this far then sleep forever
    SLEEPFOREVER;
}

```

Figure 7: Example Co-simulation Test Program

12.2 RISC-V instruction set simulator

The first case above gives full access to the API to write new test code to drive the address bus transactions. A more useful ability might be to run a complex C++ model and have this access logic components in the simulation. The test code in CoSim/tests/iss hooks up a RISC-V instruction set simulator. This is a pre-existing open-source model available on [github](https://github.com/riscv/riscv-iss) which models a RISC-V processor to the RV32GC+Zicsr standard, with machine level privileges. The model can be called from an external function and provides a means to register a callback function to be called whenever the processor does a load or store access. This callback function can choose to process the access or hand it back to the ISS. The model can also be connected to a gdb debugger as a remote target over TCP/IP and thus an IDE such as Eclipse. The test environment now looks like the following:

OSVVM's Co-simulation Framework

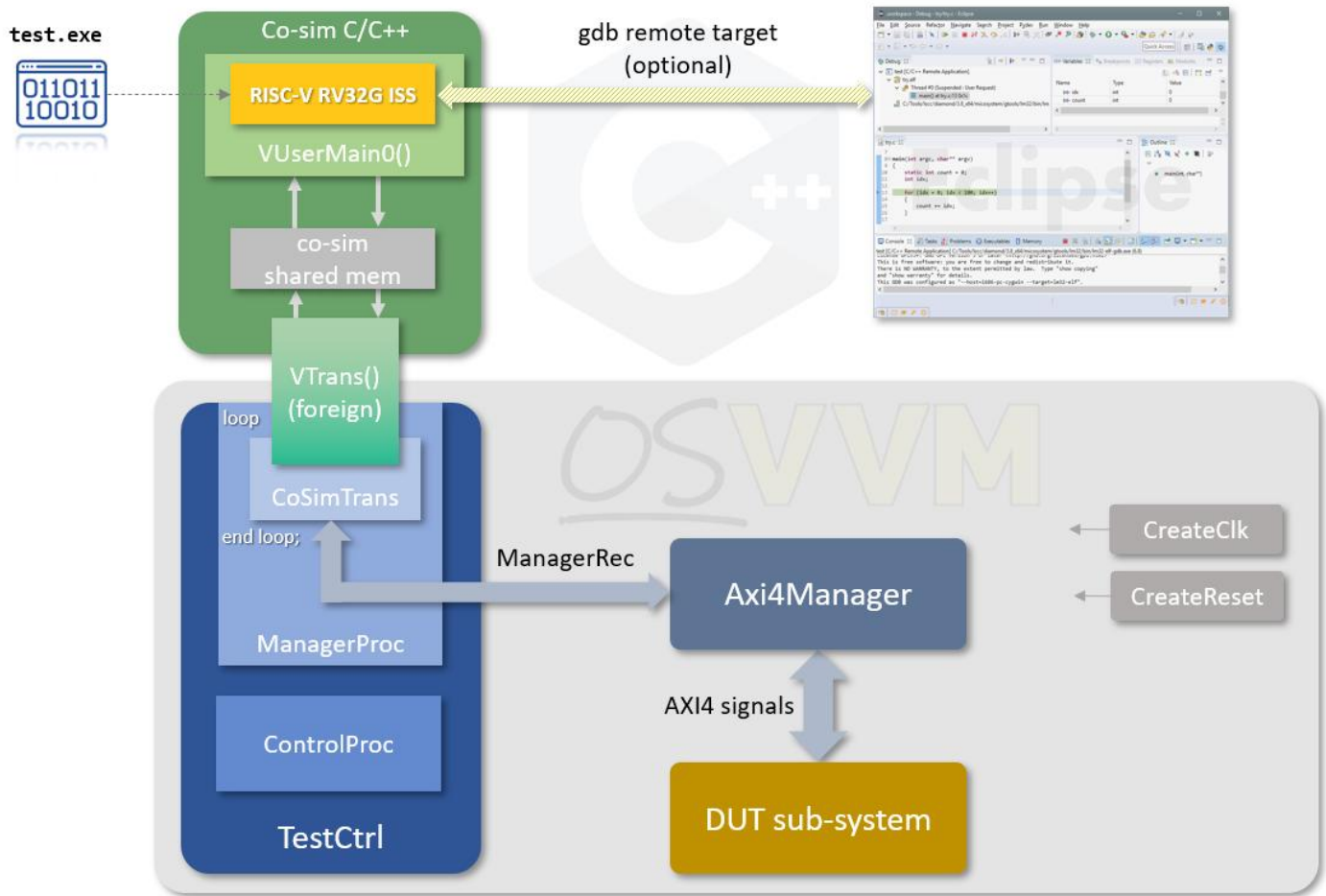


Figure 8: ISS model Co-Simulation Environment

In this test the driving of the transactions now come from within the model itself, with a pre-compiled RISC-V program (**test.exe**, in the test directory) that is one of RISC-V International's unit test—to test the **sb** (store byte) instruction. The **VUserMain0** function is now just a means to instantiate the model, load a program (if not loaded over gdb) and run it. Binary libraries for the RISC-V ISS model are provided in **CoSim/lib** and the headers in **CoSim/include**.

A simplified VUserMain0 program is shown below.

```
#include "OsvvmCosim.h"
#include "rv32.h"
#include "rv32_cpu_gdb.h"

extern "C" void VUserMain0()
{
    OsvvmCosim cosim(node);

    rv32i_cfg_s cfg; // ISS config structure
    bool error = false;

    rv32* pCpu = new rv32(); // ISS object

    // Register memory access callback
    pCpu->register_ext_mem_callback(memcosim);

    // Load a program and run the ISS model
    if (!pCpu->read_elf("test.exe")) {

        pCpu->run(cfg);
        error = check_exit_status(pCpu);

    }
    else
        error = true;

    // Clean up
    delete pCpu;

    // Flag to sim that the test is finished
    cosim.tick(10, true, error);

    // Let the simulation free run
    SLEEPFOREVER;
}
```

Figure 9: Main Program with RISC-V ISS

The registered call back function (memcosim in the example above) is called each load and store from the model and has parameters for byte address data and type. A simplified version of the callback is shown below

```
int memcosim (const uint32_t byte_addr, uint32_t &data,
              const int      type,      const rv32i_time_t time)
{
    OsvvmCosim cosim(node);
    int        cycle_count = 5;
    uint8_t    rdata8;
    uint16_t   rdata16;
    uint32_t   rdata32;

    switch (type)
    {
        case MEM_WR_ACCESS_BYTE : cosim.transWrite(byte_addr, (uint8_t)data);
            break;
        case MEM_WR_ACCESS_HWORD: cosim.transWrite(byte_addr, (uint16_t)data);
            break;
        case MEM_WR_ACCESS_WORD:  cosim.transWrite(byte_addr, (uint32_t)data);
            break;
        case MEM_WR_ACCESS_INSTR: cosim.transWrite(byte_addr, (uint32_t)data);
            break;
        case MEM_RD_ACCESS_BYTE:  cosim.transRead(byte_addr, &rdata8); data=rdata8;
            break;
        case MEM_RD_ACCESS_HWORD: cosim.transRead(byte_addr, &rdata16); data=rdata16;
            break;
        case MEM_RD_ACCESS_WORD:  cosim.transRead(byte_addr, &rdata32); data=rdata32;
            break;
        case MEM_RD_ACCESS_INSTR: cosim.transRead(byte_addr, &rdata32); data=rdata32;
            break;
        default: cycle_count = RV32I_EXT_MEM_NOT_PROCESSED;
            break
    }
    return cycle_count;
}
```

Figure 10: ISS Callback Function

This example serves to demonstrate the simplicity of connecting an existing C++ model environment to the OSVVM co-simulation features. The test bench uses the TbAxi4Memory VC as a target, with the program loaded and then run from that memory, but this just stands in for simulated DUT logic, either a peripheral or a sub-system, say. The callback might be enhanced to do some address decoding to only forward to OSVVM relevant accesses, and hand back those out of range for the rest of the model to handle.

The method described here relies on having a model that is callable from the VUserMain0 program. The simulators, in general, are executables and are not callable from external programs. The OSVVM co-simulation code provides the means to have a ‘free running’ program that can call API methods to initiate activity in the logic simulation but, ultimately, this was all initiated from the simulator process. In order to have a program, separate from the simulator, drive the transactions another method is required.

12.3 Connecting to External Programs

If the modelling system to be hooked to OSVVM is, itself, an executable and can’t be called from the co-simulation code, then a means is needed to communicate between the external model and the OSVVM co-simulation program. In CoSim/code is some source code that defines an OsvvmCosimSkt class which acts as a TCP/IP socket server. It defines a protocol for sending word or sub-word read or write

transactions, sending back responses to a connected client. The protocol is based on gdb's remote target protocol, but the class is defined in such a way as to allow a parsing and response methods to be overwritten in a derived class to alter or extend that protocol, so long as the basic structure is a start-of-packet (SOP) byte, data bytes, end-of-packet byte, followed by a fixed number of bytes (which can be 0). The current protocol is restricted to word and sub-word transactions and has no support for time advancement or interrupts, but this is easily added by extending the class to enhance the protocol and add burst transfers, time, and interrupt support. It serves as a demonstration and, of course, a user can write their own socket code. The test environment now looks like that shown in the figure below:

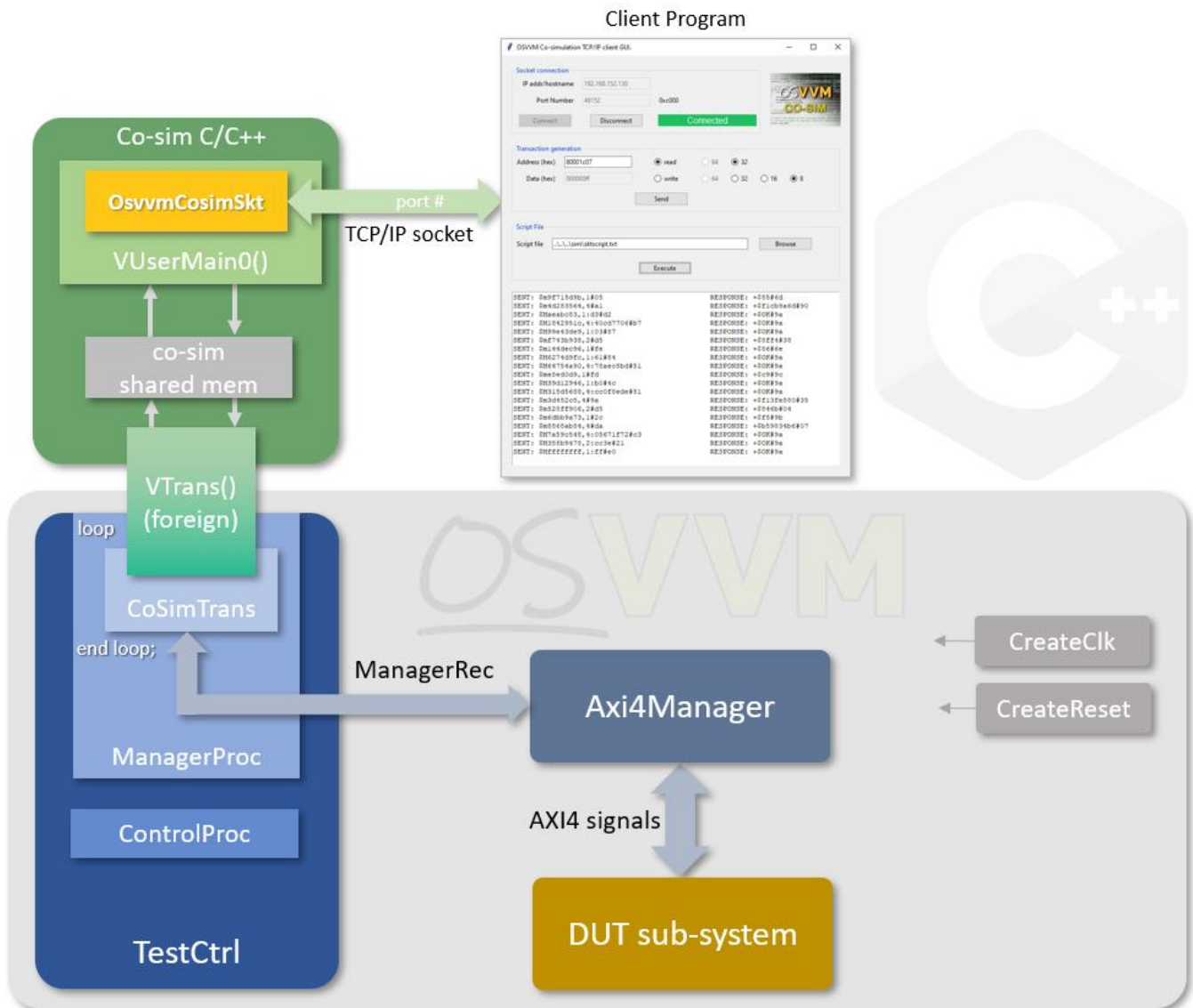


Figure 11: TCP/IP Socket Based Test Environment

The CoSim/tests/socket test code makes use of this class to open up a TCP/IP port and listen for an external connection. The VUserMain0 program is shown below:

```
extern "C" void VUserMain0()
{
    OsvvmCosim    cosim(node);
    OsvvmCosimSkt skt(node);
    bool error = false;

    if (skt.ProcessPkts() != OsvvmCosimSkt::OSVVM_COSIM_OK)
    {
        fprintf(stderr, "***ERROR: socket exited with bad status\n");
        error = true;
    }
    else
    {
        printf("DONE\n");
    }

    // Flag to the simulation we're finished, after 10 more iterations
    cosim.tick(10, true, error);

    SLEEPFOREVER;
}
```

Figure 12: Use of OsvvmCosimSkt Class

Standing in for an external program, a python script is provided in CoSim/Scripts. A batch version, used in running the demo tests is called `client_batch.py`. A GUI version (`client_gui.py`) can be used to interact directly with the simulation. When the simulation is started the simulation will wait on connection to a port (by default 49152—which is 0xC000). The GUI can then connect to this port and send commands and receive responses. The GUI is shown as the client in Figure 12 above. As well as sending and receiving data manually, the GUI can read a script, and the ISS demo test generates a script as it runs that can be used. There is also a `sktscript.txt` file in the CoSim/tests/socket directory, used in the demo tests.

12.4 InterruptHandler VC usage

In the 2022.11 release of OSVVM an Interrupt Handler VC component was added to allow for testing of the interrupt generation features of a DUT (see [2] for details). The test environment adds to the basic environment by having an additional `InterruptProc` process that can generate transactions in the same manner as a `MangerProc` process. The `InterruptHandler` arbitrates between the two sources to forward transactions to the `Axi4Manager` VC based on the state of the DUT's interrupt request line. To use the handler with co-simulation code, the multi-node capabilities are used, where the manager process uses node 0 (the case in the previous examples, and the interrupt process uses node 1. Each node will call its respective top level functions—`VUserMain0` and `VUserMain1`—each running as separate programs. This arrangement is shown in the diagram below.

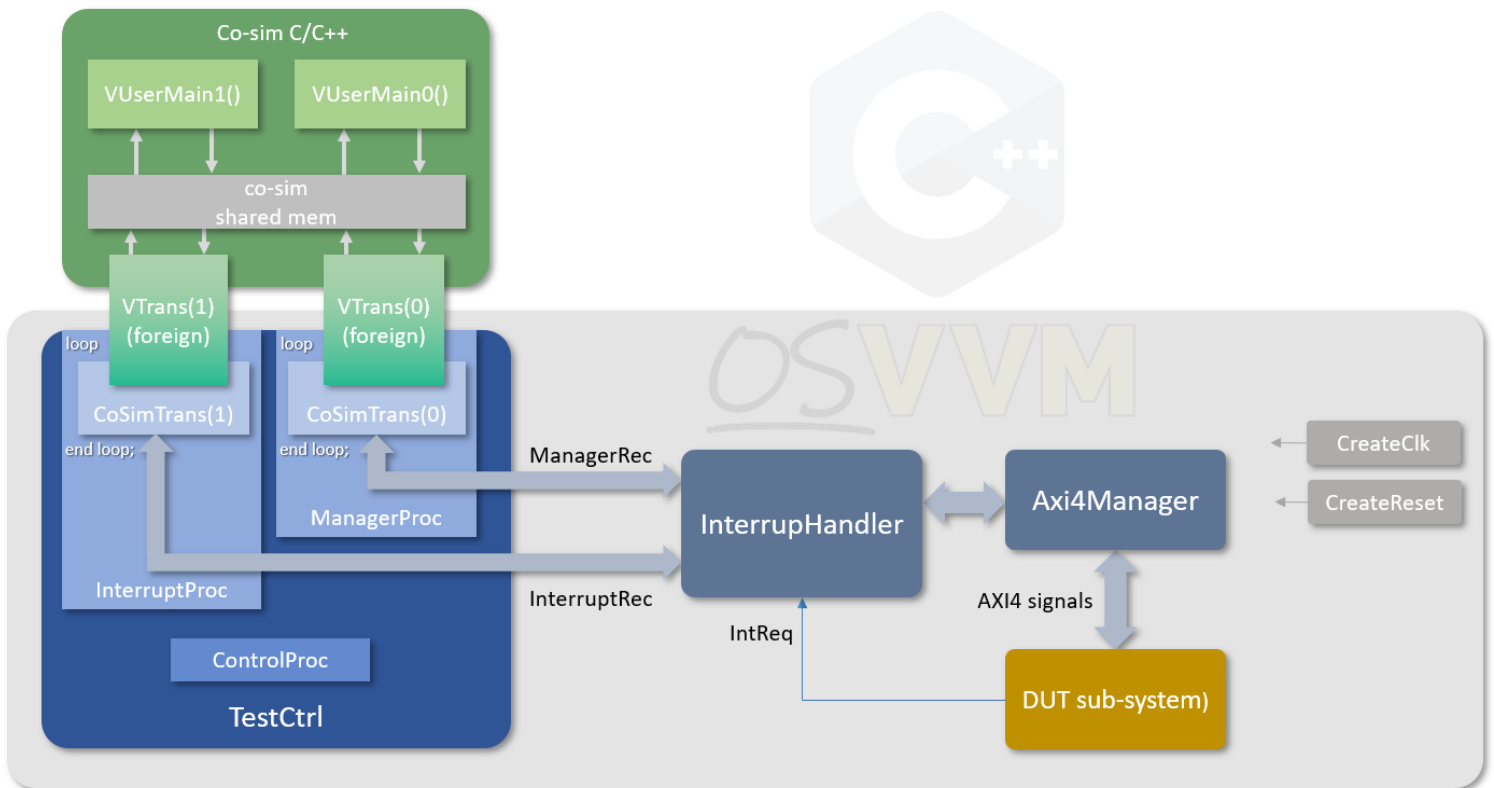


Figure 13: Co-simulation with InterruptHandler VC

A test example is given in the code in `CoSim/tests/interrupt` and runs on the `TbAxi4Memory.vhd` test bench in `Common/TbInterrupt/testbench`. The VHDL test case is `TbAxi4_InterruptCosim1.vhd` under `Common/TbInterrupt/TestCases`. The two `VUserMain` programs running simply emulate the functionality in the `TbAxi4_Interrupt1.vhd`, just using calls to the API to instigate the transactions.

12.5 Interrupt callback

The interrupt callback method of handling interrupts make use of a VHDL global signal, `gIntReq`, defined in the `InterruptGlobalSignalPkg`. The `CoSimInterruptHandler` reflects the state of the `IntReq` input and sets the global signal accordingly. The manager process of the test case can then access this signal to set the interrupt input of `CoSimTrans` as required. In the case of the `TbAxi4_InterruptCosim3.vhd` test case, the input to `CoSimTrans` is set to 1 whenever the `gIntReq` changes state. The test code in `CoSim/tests/interruptIss` registers an interrupt callback function with the co-simulation software and this function toggles a local static variable, `IntReq`, each time it is called. The test uses the RISC-V ISS model, and another function is registered as a callback from this model to inspect `IntReq` and return the status to the model. The associated RISC-V test program has the exception vectors set up and enables interrupts before running code to write to a given address to initiate an external interrupt, which the test bench monitors for and sets the `IntReq` output from the `TestCtrl_e` test bench component in the logic simulation. The RISC-V interrupt routine clears the interrupt (writes 0 to the same address that set the interrupt signal) and sets a RISC-V register to say it has been called, which the main test program then checks for, flagging a pass/fail condition.

This setup is shown in the diagram below:

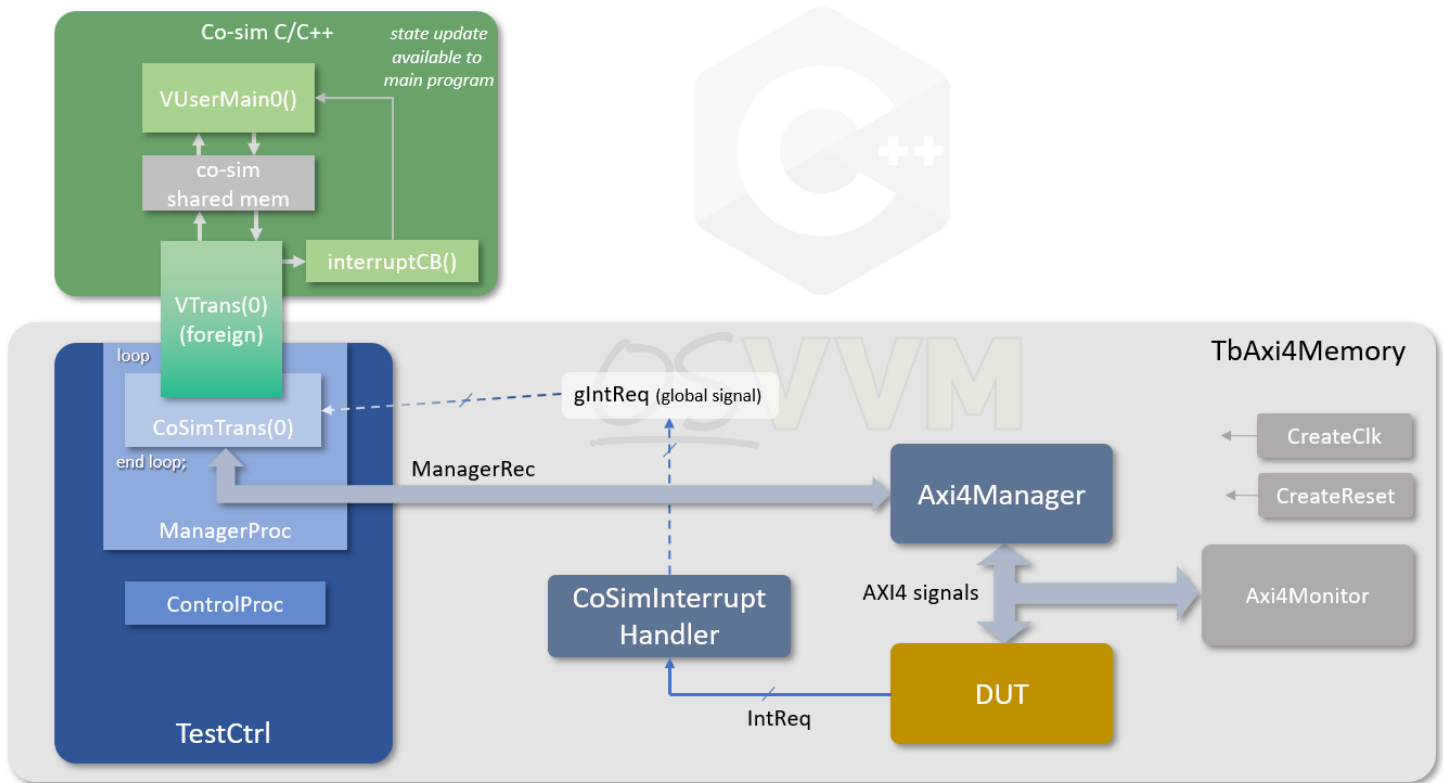


Figure 14: Interrupt Callback Environment

This test, then, demonstrates the connection between an interrupt signal in logic simulation causing an external interrupt initiating a jump to exception code, within a software processor system model.

12.6 Using GHDL callable Simulation

The default behaviour of the co-simulation code is to call one or more user programs, running in threads, with entry points at `VUserMainX` (one for each active node). The user test code is called from there and can be any code complexity, modelling whole systems if required. The assumption here is that the user's code, modelling their system, is callable from external code. This is true of the example with the RISC-V ISS, which is instantiated and run from a `VUserMain` program. This may not always be possible for pre-existing modelling systems, where the tool being used is an executable in its own right, and custom models are added to this environment and are called from there.

The supported GHDL simulator has a documented feature whereby the simulator can be called from an external program via the `ghdl_main` function. As documented, this is meant to be called from user code compiled as a shared object and loaded as such as for foreign language routines. Again, since a user's model may not be in the form of a shared object, a different model is needed to make calling the simulation possible. GHDL's two steps for compiling code is to first analyse all the VHDL files and then,

secondly, elaborate the top level. The GHDL tool must be compiled for an LLVM backend (though `gcc` might work, but not tried) and not `mcode` (see [here](#)). The first step results in a set of objects for each component in a respective library. For OSVVM, all these files are in subdirectories of `VHDL_LIBS`. The elaboration step results in an object in the compile directory with the form `e~<name of top level>.o`, for example `e~tbaxi_cosim.o`. It also results in an executable which is the simulation that can be run, but this is not yet suitable. These objects in `VHDL_LIBS` can all be gathered into a single static library, along with the relevant objects from GHDL for the normal VHDL libraries (e.g. `<root to GHDL libs>/std/08/*.o`) to be used in compiling with the main code.

The OSVVM co-simulation code itself must be compiled slightly differently so that it does not attempt to start the threads containing the `VUserMainx` code. This is done by compiling with `-DDISABLE_VUSERMAIN_THREAD` in the `gcc/g++` flags. This will still create the usual `VProc.so` and `VUser.so` shared object files. It also prevents the user code from setting the `done` argument to `true` in any call to the `tick` method. A call to this method with `done` set to `true` will block forever as it will terminate the simulation. The co-simulation software expects a response for each API call from the simulator that is no longer forthcoming, as the simulation has terminated. When the code is run as a thread from the simulator, this is not an issue, as the simulator terminating is the end of the simulation and the program exits. When called from an external program, it must not hang on this call, and external means are required to call the `tick` method with `done` set to `true` (in code which hasn't been compiled with `-DDISABLE_VUSERMAIN_THREAD`).

The two co-simulation shared objects, along with the new static library of component object files and the elaboration object can now be compiled with the user's main program to replace the original executable with one that now has user code that can call the GHDL simulator and use the co-simulation API directly. However, the simulator must be started first via the call to `ghdl_main`. This is a blocking call and won't exit until the simulation has finished. Therefore this must be run in a separate thread. Once started, and the simulation has initialised the co-simulation code, the user code can then make calls to the co-simulation API, via an `OSvvmCosimClass`. The diagram below summarises this environment

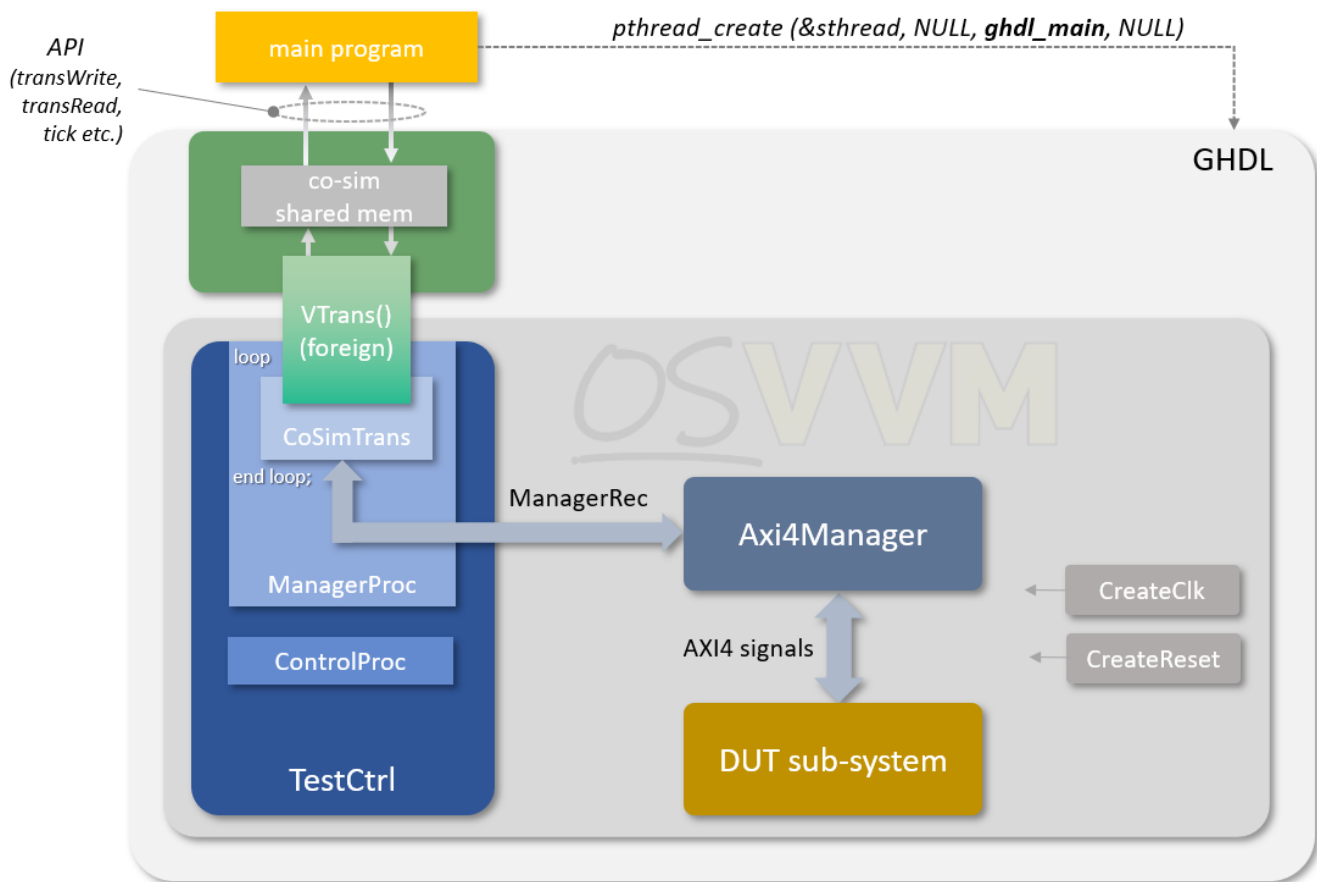


Figure 15: Callable GHDL test environment

The user code must take care of initialising the simulator, running `ghdl_main` in a thread, and waiting for it to be ready for API accesses. When the simulation finishes, the simulation must be stopped with a call to the `tick` method with the `done` argument set to `true`. As mentioned above, such a call to this method will block forever, and so this needs to be called from a separate thread. When called, the `ghdl_main` function will return and the thread it's running in will terminate. The main program can wait on this thread in order to ensure the simulation has, in fact, finished, using a `pthread_join` call. The abbreviated code fragment below summarises the use of these features.

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```
#include <pthread.h>
#include "OsvvmCosim.h"

extern "C" int ghdl_main (int argc, char **argv);
extern "C" int VUserMain0 ();

static int    node = 0;
static int    largc;
static char** largv;

// -----
int run_sim (int dummy) {
    int status = ghdl_main(largc, largv);
    return status;
}
// -----
int stop_sim_thread (int dummy) {
    OsvvmCosim cosim(node);

    // This call will hang because the simulation will not return
    // a response since it was stopped.
    cosim.tick(1, true, false);
    return 0;
}
// -----
void stop_sim(void) {
    pthread_t stop_thread;
    pthread_create(&stop_thread, NULL, (pthread_func_t)stop_sim_thread,
                  (void *)((long long)node));
}
// ===== M A I N =====
int main (int argc, char **argv) {

    OsvvmCosim cosim(node);
    pthread_t sim_thread;

    // Export arguments for use by thread code
    largc = argc; largv = argv;

    // Create a thread for calling GHDL simulator
    pthread_create(&sim_thread, NULL, (pthread_func_t)run_sim, (void *)((long long)node));

    // Allow the simulation's co-simulation code to initialise and be ready
    // for the first API call.
    cosim.waitForSim();

    // Do some tests, calling the VUserMain0 code directly.
    VUserMain0();

    // Post a done status to the simulator.
    stop_sim();

    // Wait for the simulator to exit and the thread to complete.
    pthread_join(sim_thread, NULL);

    return 0;
}
```

The steps necessary to set up the GHDL callable simulation are summarised below.

- Analyse VHDL files: `ghdl -a <list of files>`
- Elaborate top level: `ghdl -e <top level>`
- Compile co-sim code: `make -C <path to OsvvmLibraries/CoSim> [options]`
 - The `USRFLAGS` must be set to include `-DDISABLE_VUSERMAIN_THREAD`
- Compile user code: `gcc -c main.c -o main.o`
- Gather analysed file objects, user code, and GHDL library objects into a library:
 - `ar crs libtb.a main.o \`
 ``find VHDL_LIBS -name *.o` \`
 ``find <GHDL rootdir>/std/08; -name *.o` \`
 ``find <GHDL rootdir>/ieee/08; -name *.o``
- Generate a new executable:
 - `gcc ${GHDLFLAGS} e~tbaxi_cosim.o VProc.so VUser.so libtb.a \`
 `-o tbaxi4_cosim.exe`
 - The `GHDLFLAGS` are (on MSYS2/mingw-w64):
 `<GHDL rootdir>/libgrrt.a \`
 `-ldbghelp \`
 `-L. \`
 `-L<root>/lib/gcc/x86_64-mingw32/12.2.0/adalib`

The above steps, not including the analysis of the VHDL files are encapsulated in the procedure `MkVprocGhdlMain`, which has the same arguments as `MkVproc` described in section **Error! Reference source not found.** This makes use of a make file, `makefile.ghdl`, in `OsvvmLibraries/CoSim`. This make file is customisable with user overridable variables, as documented in the file's header, so can be used for user code outside of the `OsvvmLibraries` directory structure.

13 Summary

This document defines the features of the co-simulation capabilities of OSVVM. A sub-set of the Address Bus Model Independent Transaction and Stream Model Independent interfaces are exposed within a C++ environment to allow reads and writes of words or bursts, driving an `AddressBusRecType` or `StreamRecType` signal to initiate transactions on one of the pre-existing Address Bus or streaming virtual components. A single point of entry from VHDL is given with the `CoSimTrans` or `CoSimStream` procedure (for a given node), abstracting away the details of the programming interfaces of the different simulators, and keeping the OSVVM environment intact. Support for interrupts using the `InterruptHandler` VC component is also given for two usage models.

With the domain crossed from VHDL to C++ the possibilities for modelling systems that are partially simulated in C++ and partially simulated in HDL in a logic simulator is demonstrated. Two methods were described (with examples) with a model's code called directly from user co-simulation code, or a TCP/IP socket link is established to drive the simulator from an external program.

13.1 Summary of OsvvmCosim Transaction Methods

The table below summarises the transaction methods of the `OsvvmCosim` class. All of these methods are inherited (and some overloaded) by the `OsvvmCosimInt` interrupt class.

method	function
transWrite	Blocking write byte, hword, word or dword
transWriteAsync	Non-blocking write byte, hword, word or dword
transWriteAddressAsync	Non-blocking write address (split transaction)
transWriteDataAsync	Non-blocking write byte, hword, word or byte (split transaction)
transWriteAndRead	Blocking write and read byte, hword, word or dword
transWriteAndReadAsync	Non-blocking write and read byte, hword, word or dword
transBurstWrite	Blocking burst write
transBurstWriteAsync	Non-blocking burst write
transBurstWriteIncrement	Blocking burst write of incrementing pattern
transBurstWriteIncrementAsync	Non-blocking burst write of incrementing pattern
transBurstWriteRandom	Blocking burst write of incrementing pattern
transBurstWriteRandomAsync	Non-blocking burst write of incrementing pattern
transRead	Blocking read byte, hword, word or dword
transReadAddressAsync	Non-blocking read address (split transaction)
transReadData	Blocking data only read of byte, hword, word or dword (split transaction)
transTryReadData	Non-blocking data only read of byte, hword, word or dword (split transaction)
transReadCheck	Blocking read and check of byte, hword, word or dword
transReadDataCheck	Blocking data only read and check of byte, hword, word or dword (split transaction)
transTryReadDataCheck	Non-blocking data only read and check of byte, hword, word or dword (split transaction)
transBurstRead	Blocking burst read
transBurstReadCheckData	Blocking burst read and check of data (split transaction)

transBurstReadCheckIncrement	Blocking burst read and check data incrementing pattern(split transaction)
transBurstReadCheckRandom	Blocking burst read and check random pattern (split transaction)
transBurstCheckData	Blocking check of received data
transBurstCheckIncrement	Blocking check of received incrementing pattern
transBurstCheckRandom	Blocking check of received random pattern
transPushData	Blocking push of data to fifo for transmission or check
transPushIncrement	Blocking push of incrementing pattern to fifo for transmission or check
transPushRandom	Blocking push of random pattern to fifo for transmission or check
transPopData	Blocking pop of data from receive fifo
transGetTransactionCount	Get the number of interactions over the interface
transGetWriteTransactionCount	Get the number of write transactions over the interface
transGetReadTransactionCount	Get the number of read transactions over the interface
transWaitForTransaction	Wait for a transaction
transWaitForWriteTransaction	Wait for a write transaction
transWaitForReadTransaction	Wait for a read transaction

13.2 Summary of OsvvmCosimResp Response Methods

The table below summarises the response methods of the OsvvmCosimResp class.

method	function
respGetWrite	Get write transaction address and data
respTryGetWrite	Get write transaction address and data or return false if none
respGetWriteAddress	Get write address
respTryGetWriteAddress	Get write address or return false if none

respGetWriteData	Get write data
respTryGetWriteData	Get write data or return false if none
respSendRead	Send read response for received transaction
respTrySendRead	Send read response for received transaction or return false if none
respGetReadAddress	Get read address
respTryGetReadAddress	Get read address or return false if none
respSendReadData	Send response data for received transaction
respSendReadDataAsync	Send response data for received transaction asynchronously
respWaitForTransaction	Wait for a transaction
respWaitForWriteTransaction	Wait for a write transaction
respWaitForReadTransaction	Wait for a read transaction
respGetTransactionCount	Get the number of interactions over the interface
respGetWriteTransactionCount	Get the number of write transactions over the interface
respGetReadTransactionCount	Get the number of read transactions over the interface

13.3 Summary of OsvvmCosimStream Transaction Methods

The table below summarises the transaction methods of the `OsvvmCosimStream` class.

method	function
streamSend	Blocking send of byte, hword, word or dword
streamSendAsync	Non-blocking send of byte, hword, word or dword
streamBurstSend	Blocking send of burst data
streamBurstSendAsync	Non-blocking send of burst data
streamBurstSendIncrement	Blocking send of burst incrementing pattern
streamBurstSendIncrementAsync	Non-blocking send of burst incrementing pattern

streamBurstSendRandom	Blocking send of burst random pattern
streamBurstSendRandomAsync	Non-blocking send of burst random pattern
streamGet	Blocking get of byte, hword, word or dword
streamGetBurstGet	Blocking get of burst data
streamCheck	Blocking get and check of byte, hword, word or dword
streamBurstCheck	Blocking get and check of burst data
streamBurstCheckIncrement	Blocking get and check of burst incrementing pattern
streamBurstCheckRandom	Blocking get and check of burst random pattern
streamTryGet	Non-blocking get of byte, hword, word or dword
streamTryCheck	Non-blocking get and check of byte, hword, word or dword
streamBurstTryGet	Non-blocking get of burst data
streamBurstTryCheck	Non-blocking get and check of burst data
streamBurstTryIncrement	Non-blocking get and check of burst incrementing pattern
streamBurstTryRandom	Non-blocking get and check of burst random pattern
streamBurstPushData	Push data to fifo for transmission
streamBurstPushCheckData	Push data to fifo for checking against received data
streamBurstPushIncrement	Push incrementing pattern to fifo for transmission
streamBurstPushCheckIncrement	Push incrementing pattern to fifo for checking against received data
streamBurstPushRandom	Push random pattern to fifo for transmission
streamBurstPushCheckRandom	Push random pattern to fifo for checking against received data
streamGetRxTransactionCount	Get number of completed transactions for transmit interface
streamGetTxTransactionCount	Get number of completed transactions for receive interface
streamWaitForRxTransaction	Wait for a transaction completion on transmit interface
streamWaitForTxTransaction	Wait for a transaction completion on receive interface

14 About the OSVVM

The OSVVM utility and verification component libraries were developed and are maintained by Jim Lewis of SynthWorks VHDL Training and the co-simulation libraries/features were developed and are maintained by Simon Southwell. These libraries evolved from methodology and packages developed for SynthWorks' VHDL Testbenches and verification class.

Please support OSVVM by purchasing your VHDL training from SynthWorks.

15 About the Authors and Contributors

15.1 About the Co-simulation Contributor – Simon Southwell

Simon Southwell has thirty plus years of embedded software, ASIC and FPGA design experience in fields that include high performance computing, wireless, cellular modem (LTE) and processor system modelling. Mr. Southwell has developed and successfully deployed co-simulation techniques at a variety of companies using his own open-source IP.

Mr. Southwell is currently developing open source IP in areas such as RISC-V and co-simulation, is actively mentoring undergraduate and new graduate engineers in digital systems design and is also collaborating on the development of OSVVM.

If you find bugs the co-simulation packages or would like to request enhancements, Mr. Southwell can be reached at simon.southwell@gmail.com.

15.2 About the Author - Jim Lewis

Jim Lewis, the founder of SynthWorks, has thirty plus years of design, teaching, and problem solving experience. In addition to working as a Principal Trainer for SynthWorks, Mr. Lewis has done ASIC and FPGA design, custom model development, and consulting.

Mr. Lewis is chair of the IEEE 1076 VHDL Working Group (VASG) and is the primary developer of the Open Source VHDL Verification Methodology (OSVVM.org) packages. Neither of these activities generate revenue. Please support our volunteer efforts by buying your VHDL training from SynthWorks.

If you find bugs these packages or would like to request enhancements, you can reach me at jim@synthworks.com.

16 References

- [1] Address Bus Model Independent Transactions User Guide
- [2] Interrupt Handler User Guide
- [3] Stream Model Independent Transaction User Guide